Vishweshwar Tyagi

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Research Interests

Bayesian and variational inference, Monte Carlo methods, probability and stochastic processes, analysis and partial differential equations, statistical machine learning

Education

Columbia University — New York, NY
MS Data Science – GPA 3.97/4 — Sep 2021 – Dec 2022
Courses: Machine Learning, Deep Learning, Statistical Inference, Algorithms, Computer Systems
Teaching Assistant: Analysis and Optimization, Applied Machine Learning, Deep Learning, Reinforcement Learning

IIT Kanpur — Kanpur, India
MS Mathematics – GPA 9/10 — Aug 2019 – May 2021
Courses: ODE, PDE, Differential Geometry, Topology, Complex Analysis, Probabilistic Programming, Statistical NLP
Honors: Academic Excellence Awards 2019, 2020

University of Delhi — New Delhi, India
BS Mathematics – GPA 9.41/10, Math GPA 9.61/10 — Aug 2016 – May 2019
Courses: Multivariate Calculus, Numerical Methods, Series of Functions, Probability and Statistics

Research Experience

Data Scientist — Jan 2023 – Present
Movement Recovery Lab, Department of Neurology, Columbia University — New York, NY

- Developed hierarchical Bayesian model to estimate neural recruitment curves from sparse brain and spinal stimulation data, improving estimation accuracy
- Implemented mixture model to identify and downweight outliers without discarding data, improving robustness
- Developed Bayesian mixed model to detect change in motor threshold, reducing sample size to assess treatment effects in intervention studies
- Developed Bayesian adaptive design to acquire neural recruitment curves using Expected Information Gain, improving acquisition efficiency
- Optimized adaptive design speed in JAX and deployed it using Lab Streaming Layer for online experiments

Publications

(Preprint) 2026. *Bayesian adaptive design for neural recruitment curves.*
Tyagi, V., Dukkipati, S. S., Twili, D., Mikhaeil, J. M., Gelman, A., Carmel, J. B., McIntosh, J. R. arXiv. https://doi.org/10.48550/arXiv.2610.15000

2025. *Hierarchical Bayesian estimation of motor-evoked potential recruitment curves yields accurate and robust estimates.*
Tyagi, V., Murray, L. M., Asan, A. S., Mandigo, C., Virk, M. S., Harel, N. Y., Carmel, J. B., McIntosh, J. R. Brain Stimulation. https://doi.org/10.1016/j.brs.2025.09.008

Industry Experience

Data Science Intern — Summer 2022
Quartet Health — New York, NY

- Fine-tuned large language model BERT on clinical notes to identify high-risk patients for mental health conditions, increasing F2-score by 13% over XGBoost baseline
- Built data pipeline on Amazon Redshift using dbt and SQL to automate transformation of medical claims data, added unit tests to validate pipeline output
- Utilized data pipeline to evaluate insurance network quality and identify network gaps, reducing claim denial rates by 7% through outlier detection and saving $20K in referral costs

Software

`onMEP` Online acquisition of neural recruitment curves (https://github.com/hbmep/onmep)

`hbMEP` Hierarchical Bayesian estimation of neural recruitment curves (https://hbmep.github.io/hbmep/)

Teaching

Teaching Assistant, Columbia University

- Applied deep learning (CS W4995) Fall 2022
- Applied machine learning (CS W4995) Spring 2022
- Analysis and optimization (MATH V2500) Spring 2022
- Reinforcement learning (EE E6885) Fall 2021
- Calculus I (MATH UN1101) Fall 2021

Technical Skills

Programming: Python, R, C/C++, SQL (Postgres)

Probabilisic Programming: NumPyro, Pyro, Stan

Machine Learning: JAX, scikit-learn, XGBoost, Hugging Face, OpenAI

Big Data: dbt, Spark, BigQuery, Redshift

Developer Tools: Git, Bash, Linux, SSH, Vim, VS Code, Docker, Google Cloud Platform

Conferences

Talk: *Hierarchical Bayesian estimation of motor-evoked potential recruitment curves yields accurate and robust estimates.* Society for Neuroscience (SfN) Nanosymposium on Analytical Computational Tools, October 2024, Chicago, Illinois.

Poster: *Hierarchical Bayesian estimation of motor-evoked potential recruitment curves yields accurate and robust estimates.* 11th Annual Minnesota Neuromodulation Symposium – Neuromodulation of Spinal Cord Injury: Translational Opportunities, April 2024, Minneapolis, Minnesota. 3rd Place Poster Award.